DTC P0A8F:00 [SKYACTIV-D 1.5]

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Note
- To determine the malfunctioning part, proceed with the diagnostics from "Function Inspection Using M-MDS".

Details On DTCs

DESCRIPTION	Power system: Low input	
DETECTION CONDITION	Determination conditions	• Any one of the following conditions is met: — Battery voltage is **11 V or less** when ignition is switched off after leaving vehicle for **30 days or more**. — Battery voltage is **11 V or less** for a continuous **5 days** when ignition is switched off.
	Preconditions	• Difference in battery voltage for **30 min** since ignition was switched off is specified value or less • Electrical current from battery for **30 min** since ignition was switched off is within specified value
	Drive cycle	• **1**
	Self test type	• CMDTC self test
	Sensor used	• Current sensor
FAIL-SAFE FUNCTION	• Not applicable	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Flashes i-stop warning light (amber). • The engine cannot be started or the engine may stall due to battery voltage decrease.	
POSSIBLE CAUSE	• Battery malfunction • Current sensor malfunction • BCM malfunction • PCM malfunction	

System Wiring Diagram
- Not applicable

Function Explanation (DTC Detection Outline)
- The battery assures voltage for engine starting and while the vehicle is being driven, and for the vehicle's electrical devices when the engine is stopped by i-stop.
- The PCM determines a dendrite malfunction by the condition in which the battery voltage value continues to be low for the specified time. In this case, the PCM determines a malfunction in a battery-related part and stores a DTC.

Repeatability Verification Procedure
1. Start the engine

PID Item/Simulation Item Used In Diagnosis
- Not applicable

Function Inspection Using M-MDS

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**PURPOSE: VERIFY IF BATTERY VOLTAGE IS FALSELY RECOGNIZED BY DTC RELATED CURRENT SENSOR** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the PENDING CODE/DTC P058A:00 also present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC P058A:00 [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
3	**PURPOSE: VERIFY IF BATTERY VOLTAGE IS FALSELY RECOGNIZED BY DTC RELATED CAN OR LIN COMMUNICATION** • Perform the PCM and BCM DTC inspection using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) (See DTC INSPECTION [BCM].) • Are DTCs related CAN or LIN communication recorded?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-D 1.5].) (See DTC TABLE [BCM].) Go to the troubleshooting procedure to perform the procedure from Step 1.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure
Intention of troubleshooting procedure

- Step 1
  - — Perform a unit inspection of the battery.
- Step 2—3
  - — Verify that primary malfunction is resolved and there are no other malfunctions.

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: DETERMINE INTEGRITY OF BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is there any malfunction?	Yes	Replace the battery, then go to the next step. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
2	**PURPOSE: VERIFICATION OF VEHICLE REPAIR COMPLETION** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure.) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
3	**PURPOSE: VERIFY IF THERE IS ANY OTHER MALFUNCTION** • Is any other DTC or pending code stored?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.